

Final Project Report Template

Project: Food Ordering Behaviour and Consumer Trends | Dataset: 50,000 food orders | Tool: Tableau

1. Introduction

1.1. Project overviews

The project, "Food Ordering Behaviour and Consumer Trends", analyzes a dataset containing 50,000 food orders. The purpose is to understand customer ordering patterns using City, Cuisine, Age Group, Restaurant Type, Meal Type, Company, Order Value, Rating Given and Time Taken To Order. Tableau was used to explore the data and create visualizations, a dashboard and a story. The completed Tableau work was also integrated into a web interface using Flask and deployed as a publicly accessible project.

The final dashboard contains eight analytical visualizations: Flavors Across Cities, Analysis of Orders Across Company Type and Meal Type, Revenue Contribution by Meal Type, Order Distribution by Age Group and Restaurant Segment, Customer Rating Analysis by Meal Type, Cities By Order Value/Order Volume, Distribution of Customer Preferences across Cuisines, and Time Taken to Order. A separate Total Orders KPI is also included for an overall summary.

1.2. Objectives

To analyze 50,000 food orders and identify major ordering patterns.

To compare food preferences across cities and cuisines.

To study order distribution across age groups and restaurant segments.

To compare orders across company types and meal types.

To analyze customer ratings by meal type.

To examine city-wise order volume and meal-type contribution to total order value.

To present the findings through Tableau visualizations, a dashboard, a story, and a Flask-based web interface.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Problem Statement 1: I am a food ordering customer. I'm trying to choose suitable food and restaurants based on my preferences, meal type, cuisine and expected service. But it can be difficult to understand which cuisines, meal types and restaurant categories are most commonly preferred by customers. Because the available food ordering data contains a large number of orders with different customer and restaurant characteristics. Which makes me feel that it is difficult to identify clear trends and make informed choices.

Problem Statement 2: I am a restaurant or food-service business. I'm trying to understand customer ordering behaviour and identify the types of food and meals that receive higher demand. But it is difficult to identify important patterns from a large number of individual orders. Because customer behaviour varies

according to city, age group, cuisine, restaurant type, company type and meal type. Which makes me feel that a clear visual analysis of customer ordering behaviour is required.

2.2. Project Proposal (Proposed Solution)

The proposed solution is a Tableau-based analytical project that converts raw food-ordering records into understandable visual information. The project provides descriptive and exploratory analysis rather than a real-time recommendation system.

The approach includes data understanding, quality checking, preprocessing, aggregation, visualization, dashboard development, story creation, web integration using Flask and deployment. The solution provides a Total Orders KPI and eight analytical visualizations covering city-cuisine patterns, age-group and restaurant-segment comparison, cuisine preferences, company-meal analysis, rating analysis, city order volume, meal-type revenue contribution and order-time analysis.

2.3. Initial Project Planning

The project was planned in three sprints. Sprint 1 covered problem definition, data collection and data exploration. Sprint 2 covered preprocessing, data preparation and visualization development. Sprint 3 covered dashboard development, story preparation and final presentation.

Planned sequence: Problem Definition → Data Collection → Data Exploration → Data Preprocessing → Data Preparation → Visualization → Dashboard → Story → Flask Web Integration → Deployment → Final Presentation.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Raw data source: Food Ordering Dataset. The dataset contains approximately 50,000 food-order records with customer, restaurant, city, cuisine, meal and order-related fields. The project uses the provided/course dataset for academic analysis.

Main fields used include City, Cuisine, Age Group, Restaurant Type/Segment, Meal Type, Company, Order Id, Order Value, Rating Given and Time Taken To Order.

Dataset URL, exact file size and external source details were not provided, so no unsupported URL or file-size value has been added.

3.2. Data Quality Report

The data-quality review focused on missing values, duplicate records, inconsistent categorical values, numerical data types and invalid numerical entries.

Resolution approach: verify important fields, check Order Id records for duplicates, standardize category text where required, ensure Order Value, Rating Given and Time Taken To Order are numerical, and validate values before aggregation.

3.3. Data Exploration and Preprocessing

The dataset contains 50,000 orders. Categorical fields were treated as dimensions and numerical fields as measures. Tableau aggregations such as COUNT(Order Id), SUM(Order Value) and AVG(Rating Given) were used for analysis.

City, Cuisine, Age Group, Restaurant Type, Company and Meal Type were used as dimensions. Order Value, Rating Given and Time Taken To Order were used as measures.

The project mainly uses a single dataset, so no complex multi-table data model was required. No major column splitting or merging was required for the final dashboard.

4. Data Visualization

4.1. Framing Business Questions

1. How many total food orders are present in the dataset?
2. How are cuisine preferences distributed across different cities?
3. How are orders distributed across age groups and restaurant segments?
4. Which cuisines have higher order volumes?
5. How do orders vary across company type and meal type?
6. How do average customer ratings compare across meal types?
7. How does order volume vary across cities?
8. What percentage of total order value is contributed by each meal type?
9. What does the distribution of time taken to order show?

4.2. Developing Visualizations

Eight analytical visualizations were developed: Flavors Across Cities heat map; Order Distribution by Age Group and Restaurant Segment grouped bars; Distribution of Customer Preferences across Cuisines horizontal bars; Analysis of Orders Across Company Type and Meal Type stacked bars; Customer Rating Analysis by Meal Type bars; Cities By Order Value/Order Volume treemap; Revenue Contribution by Meal Type pie chart; and Time Taken To Order distribution. A separate Total Orders KPI was used to show the overall number of orders.

Total orders = 50,000. Total Order Value shown in the dashboard = ₹2,73,86,555.

City order counts shown in the dashboard: Mumbai 8,443; Pune 8,354; Bangalore 8,353; Delhi 8,309; Chandigarh 8,279; Hyderabad 8,262.

Average ratings by meal type: Breakfast 3.00954, Dinner 2.99169, Lunch 2.99017 and Snacks 2.98618.

5. Dashboard

5.1. Dashboard Design File

Dashboard title: "Food Ordering Behaviour and Consumer Trends". The dashboard uses a structured grid layout with clear borders between sections.

The dashboard contains the Total Orders KPI and eight analytical visualizations: Flavors Across Cities; Analysis of Orders Across Company Type and Meal Type; Revenue Contribution by Meal Type; Order Distribution by Age Group and Restaurant Segment; Customer Rating Analysis by Meal Type; Cities By Order Value/Order Volume; Distribution of Customer Preferences across Cuisines; and Time Taken To Order.

Dashboard observations: 1) 50,000 total orders were analyzed. 2) Order volumes are fairly balanced across the six cities, with Mumbai and Pune among the higher-volume cities. 3) Desserts and Chinese

cuisine show strong customer preference based on displayed order counts. 4) Order distribution varies across age groups and restaurant segments, with Gen Z and Millennials contributing large volumes. 5) Meal types have nearly equal revenue contributions and very similar average ratings.

6. Report

6.1. Story Design File

Story title: "Executive Summary: Food Ordering Behaviour & Consumer Trends". The story presents the findings in a sequence covering city and cuisine patterns, customer segments, company and meal patterns, ratings, order volume and revenue contribution.

Story points: 1) Executive Overview – overall order volume and project scope. 2) City & Cuisine – cuisine patterns across cities. 3) Age & Restaurant Segments – comparison of age groups with restaurant types. 4) Cuisine Preferences – order distribution across cuisines. 5) Company & Meal Type – relationship between company type and meal type. 6) Customer Satisfaction – average ratings by meal type. 7) City Order Volume – comparison of city-wise orders. 8) Revenue Share by Meal Type – contribution of each meal type to total Order Value. 9) Final Summary – key findings from the complete dashboard.

7. Performance Testing

7.1 Utilization of Data filters

The final dashboard primarily uses legends and visual selections for exploration. No dedicated user-created filter field is counted in the final dashboard.

7.2 No of Calculation Field

Calculation fields created specifically by the project: 0. The analysis uses standard Tableau aggregations such as COUNT(Order Id), SUM(Order Value) and AVG(Rating Given), rather than custom calculated fields.

7.3 No of Visualization

Number of analytical visualizations in the final dashboard: 8. In addition, a separate Total Orders KPI is included as a summary metric.

8. Conclusion/Observation

The project successfully transformed 50,000 food-order records into an interactive Tableau dashboard and story, and made the completed analysis accessible through a Flask-based web interface. The analysis compares customer ordering behaviour across cities, cuisines, age groups, restaurant segments, company types and meal types.

The dashboard shows relatively balanced city-wise order volumes, strong order activity across several cuisines, differences in age-group and restaurant-segment distribution, balanced meal-type revenue contribution, and very similar average ratings across meal types.

9. Future Scope

Add interactive filters for City, Cuisine, Age Group, Restaurant Type and Meal Type.

Add trend analysis if date/time information becomes available.

Develop predictive models for order demand, customer preferences and order value.
Add customer segmentation using machine learning techniques.
Create a recommendation system for cuisines or restaurants based on customer behaviour.
Connect the dashboard to a regularly updated data source for near-real-time monitoring.

10. Appendix

10.1. Source Code(if any)

The Tableau dashboard itself was developed using Tableau visualizations, worksheet configurations, dashboard layout and standard Tableau aggregations. In addition, Flask was used for the web integration of the completed Tableau work. The Flask application provides a web page that presents the Tableau dashboard and Tableau story in one publicly accessible interface.

The project was tested locally before deployment. For production deployment, the Flask application was deployed on Render and Gunicorn was used as the production server. The GitHub repository was connected to the deployed application.

10.1.2. Technologies Used

Python, Flask, Tableau Public, HTML/CSS, GitHub, Render and Gunicorn were used in the completed project. Tableau was used for data analysis and visualization, while Flask was used for the web interface and integration of the Tableau outputs.

Tabluecourse/

- |— README.md
- |— app.py
- |— requirements.txt
- |— food_ordering_behavior_dataset.csv
- |— templates/
 - |— index.html

The main project structure used for the web application is:

The web integration was implemented using Python and Flask. The application uses an app.py file as the Flask entry point and an HTML page in the templates folder to provide the web interface. The Tableau dashboard and Tableau story are embedded into the webpage so that the completed analysis can be accessed from a single website.

10.1.1. Flask Web Integration

10.2. GitHub & Project Demo Links

GitHub Link: [topareaaditya458-ops/Tabluecourse](https://github.com/topareaaditya458-ops/Tabluecourse)

Project Demo: [Food Ordering Behaviour and Consumer Trends](#)

Tableau Dashboard: [food ordering behaviour and consumer treands | Tableau Public](#)

Tableau Story: [Professional & clear- food ordering behaviour story | Tableau Public](#)